# Ishaan Jain

## Nanodevices & Nanotechnology Engineer | Graduate Student at KU Leuven

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Leuven, Belgium

I've always been driven to solve complex problems through scientific methods—luckily, that led me to Engineering. My expertise lies in building nanodevices from the ground up, where I apply a curious, first-principles approach to solve the complex physical and technical challenges inherent in practical device development. I thrive on the challenge of translating lab innovations into impactful technologies, dedicated to ensuring that advanced research moves from the cleanroom to the people effectively.

## Education

2024 - 2026(exp.) Master of Nanoscience, Nanotechnology and Nanoengineering with specialization in Nanodevices
KU Leuven, Belgium
**Thesis Subject :** Modeling MOL parasitics for Advance Technology CFETs at IMEC, **GPA :** Cum Laude

2020-2024 Bachelor of Technology in Engineering Physics with focus on Eletrical Engineering
Indian Institute of Technology Delhi, India
**Thesis Subject :** Designing and Simulation Single Photon Avalanche Devices, **GPA :** First Class

## Work experience - Internships

**August 2025 – July 2025** | **Student Intern | Interuniversity Microelectronics Centre | Leuven, Leuven, Belgium**
- Performed Design-Technology Co-Optimization (DTCO) by simulating 3D device structures in TCAD and generating electrical models for logic standard cells.

Synopsys TCAD | Python

**July 2023 – June 2023** | **Opto-Mechanical Instrument Engineer | Tata Institute of Fundamental Research, Mumbai, India**
- Designed an optical instrument using Solidworks CAD software.
- Conducted finite element analysis (FEA) using Ansys to ensure structural and thermal integrity.
- Achieved 16 micro-meter and 5 micro-radian precision by optimization.

Solidworks | Ansys Mechanical FEA | Zemax | MATLAB | Simulink

**July 2022 – June 2022** | **Robotics Instructor | Rancho Labs | I-Hub Foundation for Cobotics, Delhi, India**
- Designed and instructed an introductory robotics course for school students.

Arduino IDE | 3D printing | PCB designing

## Projets

**Analytical Modeling for MOL Parasitics of Advance devices - Master Thesis | IMEC** — September 2025 - Ongoing

Using diverse mathematical methods to create models for CFETs, iteratively refining by comparison with simulation results.
Built a custom script to generate and simulate variable CFET dimensions, enabling parametric exploration of geometries.
Simulated 3D structures to extract SPICE models with parasitics, driving Power-Performance-Area (PPA) trade-offs.
Authored Python-based rendering and visualization scripts to accelerate the analysis of complex device architectures.

Python | Jupyter notebook | Synopsys TCAD

**Simulation and Optimization of Optical NEMS based XOR Gate - Course Project | IMEC** — Feb 2025 - June 2025

Replicated a published device structure to optimize eletrical and dimensional parameters via simulation and analytical analysis.

COMSOL MultiPhysics | MATLAB

**Design of a simple Operational Telescopic Amplifier (OTA) - Course Project | MICAS| KU Leuven** — Sep 2024 - Nov 2024

Designed and later tested the fabricated chip for specific Gain and Bandwidth under power & area constraints.

Cadence Virtuoso

**Designing Single Photon Avalanche Diodes (SPAD) - Bachelor Thesis | Dept. of EE | IIT Delhi** — Aug 2023 - May 2024

Model physical devices such as PN junction diodes, bipolar diodes and MOSFETs to understand physics and simulations.
Conducted study of Planar and Epitaxial SPADs, scrutinizing breakdown voltage, structural modifications, and other performance parameters to increase Photon Detection Efficiency for 800nm wavelength infrared light to 30%.

Silvaco TCAD | LaTeX

**Design and Fabricating Equipment for Total Solar Eclipse Chase 2023 | QGal Group | IIT Delhi** — Sep 2022 - June 2023

Captured low-intensity images using custom 3D-printed filters for changes in solar spots.
Designed and manufactured mounts for telescopes, power banks, and camera equipment.
Applied photometry techniques on-site for data collection during the Total Solar Eclipse in Exmouth, Australia

Solidworks | 3D Printing | Electronics

**8-bit Vector Multiplication Microprocessing Chip Design - Course Project| Dept. of EE| IIT Delhi** Aug 2023 - Nov 2023

Designed behavioural code in Verilog for an 8-bit vector multiplication microprocessor unit.
Conducted RTL compilation using Cadence genus and layout generation using Innovus to perform static timing analysis and design rule checks on the layout.

Cadence genus | Innovus | Virtuoso | Verilog | SimpleRISC

**Theoretical Computation and Scientific Data Analysis | Dept. of Physics | IIT Delhi** April 2023

Utilized Julia language for data analysis and accurate numerical solutions.
Using the Crank-Nicolson method for the time-dependent Schrödinger equation and Lagrangian.

Julia | Python | Git

**Engineer at Accumulator department | Electrical Racecar - Formula student Team | IIT Delhi** May 2021 - May 2022

Designed and fabricated a 400V Lithium-Ion Pouch cell Battery module and Battery Management System for an 80kW powertrain.
Conducted detailed testing using coulomb counting to model performance using MATLAB.
Designed, debugged, and implemented safety circuits using Altium PCB Designer.
Engineered lightweight aluminium housings & charging cart using tubular CAD modelling.
Achieved 2nd positions in Best Vehicle Design, Best Powertrain, and Best Battery Design at Formula Bharat, securing an overall 3rd position.

3D Modeling | Manufacturing Process | Laser Cutting | Mechatronics | Altium PCB Designer | LTSpice

## Skills

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|---|---|
| **Tech. Languages and Frameworks :** | Python, Verilog (Basics), MATLAB, Simulink, Julia, C/C++, SimpleRISC |
| **Engineering Softwares :** | Solidworks, Ansys Mechanical, Altium PCB Designer, LTSpice, Origin, Arduino IDE |
| **Hardware :** | 3D-Printing, PCB-debugging, Soldering, Wiring, Microcontroller programming |
| **Spoken Languages :** | English (Fluent), Hindi (native), Dutch (Beginner) |
| **Other skills :** | Cross-disciplinary Thinker, Outcome-oriented, Initiative-Driven Innovator |

## Relevant CourseWork

Nanodevices and circuits
Technology of Integrated Systems, Reliability and Yield, Physical Material Characterization, MEMS and Microsystem
Undergraduate Physics
Solid state Physics, Statical Physics, Quantum Mechanics, Electrodynamics, Semiconductor Devices
Electrical Engineering
Design and Implementation of Analogue circuits, Comouter Architecture, Intro to VLSI, Physical Electronics, Signals and Systems, Digital Electronics, Design of Digital Integrated Circuts, Analog and Mixed-Signal Electronics for Signal Processing

## Scholastic Achievements

Sakura Science Exchange | Japan Science and Technology Dept. 2024
Guided visits to top universities and R&D centers in Japan to foster international collaboration (SSC ID : 24300043).
All India Rank 1756 | IIT JEE Advanced 2020
Secured rank 1756 among 1.5 million candidates in one of the world's most competitive engineering entrance exams.
Top 1% in Indian National Physics Olympiad 2019
Organized by IAPT, to select students for the International Physics Olympiad (IPhO).
Top 1% Twice in Indian National Astrophysics Olympiad 2018,2019
Organized by IAPT, to select students for the International Astronomy Olympiad (IAO), selected to compete internationally.

## Extra Curriculars

Head PoC, Physics and Astronomy Club
Initiated the university's first student-led space exploration project. Organised multiple events and competitions nationwide.
Hostel Representative to Univ. Welfare Council
Elected from 600+ peers to manage welfare, administrative tasks and organise events.
Additional activities
Won events as a member of Dramatics Club and Debate Club, as well as won multiple business case competitions during my bachelor's.

## References

*Available upon Request.*